

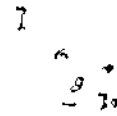
第六章 偏振

§ 1.

光 → 横波

偏振片：透振方向

马吕斯定律： $I = A^2 = \langle A_0 \cos \theta \rangle^2 = I_0 \cos^2 \theta$



偏振类型：

(1) 自然光：任何方向

$$I = \int_0^{2\pi} I_0 \cos^2 \theta d\theta = \frac{1}{2} I_0$$

(2) 线偏振光

(3) 部分偏振光

$$\text{定义偏振度 } P = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

经过偏振片： I_m, I_n 非相干叠加

以出现 I_m, I_n 方向为 x, y

$$I(\alpha) = I_m \cos^2 \alpha + I_n \sin^2 \alpha \quad \text{①: 自然光}$$

$$= \underbrace{I_m}_{\text{①}} + \underbrace{(I_m - I_n) \sin^2 \alpha}_{\text{②}} \quad \text{②: 线偏振光}$$

* 部分偏振性

$$\text{定义: 线偏振数密度 } P = \frac{\Delta N}{N_0}$$

$$I_0 = \int_0^{2\pi} i_0(\theta) d\theta \quad i_0 \text{ 为每一线偏振光的光强}$$

经偏振片：

$$I(\alpha) = \int_0^{2\pi} i_0 \cos^2(\theta - \alpha) P(\theta) d\theta$$

自然光： $P = P_0$

$$I_0 = 2\pi i_0 P_0$$

$$I = i_0 P_0 \int_0^{2\pi} \cos^2(\theta - \alpha) d\theta = \pi i_0 P_0 = \frac{1}{2} I_0$$

$$I(\alpha) = I_x \cos^2 \alpha + I_y \sin^2 \alpha + \frac{1}{2} I$$

即 $I_1 \neq 0 \Rightarrow$ 自然光分解为两个正交分量非相干叠加

部分偏振光:

$$I_p(\alpha) = \int_0^{2\pi} i_0 \cos(\theta - \alpha) \rho(\theta) d\theta = \frac{1}{2} i_0 \int_0^{2\pi} [1 + \cos(2\theta - 2\alpha)] \rho(\theta) d\theta$$

$$= I_p + I_p \cos$$

$$I_p = \frac{1}{2} i_0 \int_0^{2\pi} \rho(\theta) d\theta$$

$$I_p = \frac{1}{2} i_0 \int_0^{2\pi} \rho(\theta) \cos(2\theta - 2\alpha) d\theta$$

$$= \frac{1}{2} i_0 \int_0^{2\pi} \rho(\theta) [\cos(2\theta) \cos(2\alpha) + \sin(2\theta) \sin(2\alpha)] d\theta$$

$\exists \bar{\theta} \in [0, 2\pi)$, $\bar{\theta}' \in (0, 2\pi)$ 积分中值定理

$$= \frac{1}{2} i_0 [\rho(\bar{\theta}) \cos(2\bar{\theta}) \cos(2\alpha) + \rho(\bar{\theta}') \sin(2\bar{\theta}') \sin(2\alpha)] \cdot 2\pi$$

$$= \sqrt{K^2 + K'^2} \cos(2\alpha - \theta)$$

$$\rightarrow I_m = I_p + \sqrt{K^2 + K'^2}, I_n = I_p - \sqrt{K^2 + K'^2}$$

以最大、最小光强方向为 x, y 方向

(设 $\theta = 0$) x 轴为 I_m

$\theta = \pi$: y 轴为 I_n

$$I = I_p + \sqrt{K^2 + K'^2} \cos 2\alpha$$

$$I = I_m \cos^2 \alpha + I_n \sin^2 \alpha$$

$$= \frac{1}{2} (I_m + I_n) + \frac{1}{2} (I_m - I_n) \cos(2\alpha)$$

$$= I_m + (I_m - I_n) \cos^2 \alpha$$

$$= I_m \sin^2 \alpha + I_n \cos^2 \alpha \quad (+\Delta I)$$

$$\rightarrow \Delta I = 0$$

$\Rightarrow$ 对部分偏振光, 以出现 I_m, I_n 方向分解的交叉光是不相干的

若任意选取 (x', y')

$$I_1 = I_m \cos^2 \alpha_0 + I_n \sin^2 \alpha_0$$

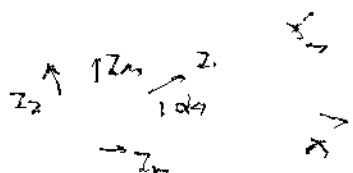
$$I_2 = I_m \sin^2 \alpha_0 + I_n \cos^2 \alpha_0$$

$$I_p(\alpha) = I_1 \cos^2(\alpha - \alpha_0) + I_2 \sin^2(\alpha - \alpha_0) + \Delta I$$

$$\rightarrow \Delta I = I_m \cos^2 \alpha + I_n \sin^2 \alpha - I_1 \cos^2(\alpha - \alpha_0) - I_2 \sin^2(\alpha - \alpha_0)$$

同样有

$$\sigma = \left| \frac{\Delta I}{I_1} \right| = \left| \frac{I_m \cos^2 \alpha + I_n \sin^2 \alpha - I_1 \cos^2(\alpha - \alpha_0) - I_2 \sin^2(\alpha - \alpha_0)}{I_1} \right|$$



$\vec{p} = p_x \hat{x} + p_y \hat{y} + p_z \hat{z}$

(4) 偏振光

$$\begin{cases} E_x = A \cos(\omega t) \\ E_y = A \cos(\omega t + \frac{\pi}{2}) \end{cases} \quad \begin{array}{l} + : \text{右旋偏振光} \\ - : \text{左旋偏振光} \end{array}$$

$$\vec{E} = A \cos(\omega t) \hat{x} + A \cos(\omega t + \frac{\pi}{2}) \hat{y}$$

$$I = A_x^2 + A_y^2 = 2A^2$$

→ 偏振片 $I: A^2$

(5) 椭圆偏振光

$$\begin{cases} E_x = A_x \cos(\omega t) \\ E_y = A_y \cos(\omega t + \delta) \end{cases}$$

$$\vec{E} = A_x \cos(\omega t) \hat{x} + A_y \cos(\omega t + \delta) \hat{y}$$

经过偏振片: x, y 方向振动叠加

$$\begin{aligned} I(\alpha) &= A_x^2 \cos^2 \alpha + A_y^2 \sin^2 \alpha + 2A_x A_y \sin \alpha \cos \alpha \cdot \cos \delta \\ &= \frac{1}{2} I_x [1 + \cos(2\alpha)] + \frac{1}{2} I_y [1 - \cos(2\alpha)] + \sqrt{I_x I_y} \sin(2\alpha) \cos \delta \\ &= \frac{1}{2} (I_x + I_y) + \frac{1}{2} (I_x - I_y) \cos(2\alpha) + \sqrt{I_x I_y} \sin(2\alpha) \cos \delta \\ &= \frac{1}{2} (I_x + I_y) + \sqrt{\frac{1}{4} (I_x - I_y)^2 + I_x I_y \cos^2 \delta} \cdot \cos(2\alpha - \varphi) \\ &= \frac{1}{2} (I_x + I_y) + \frac{1}{2} \sqrt{I_x^2 + I_y^2 + 2I_x I_y \cos 2\delta} \cdot \cos(2\alpha - \varphi) \end{aligned}$$

其中 $\varphi = \arctan \frac{2\sqrt{I_x I_y} \cos \delta}{I_x - I_y}$

→ ① 正椭圆偏振光

$$I = I_x + (I_y - I_x) \sin^2 \alpha$$

② 圆偏振光

$$I = \frac{1}{2} (I_x + I_y) = \frac{1}{2} I_0$$

* 偏振光干涉实验:

菲涅耳公式

1. 菲涅耳反射折射公式

局部坐标系: p, s, k

$$\vec{p} \times \vec{s} = \vec{k}$$

p : 平行于入射面 s : 垂直于入射面

k : 波矢

p, s 与解量 x : 特征振动

p, s 方向振动相互独立, p 光入射不会有 s 光出射

例: $\vec{E} \times \vec{H} \propto \vec{k}$

若 $H_{is} = 0, H_{rs} = 0, H_{is} \neq 0$ 则 $H_{rs} \neq 0$

$$\vec{E}_1(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k}_1 \cdot \vec{r} - \omega t + \varphi_0)} = \vec{E}_0 e^{i\vec{k}_1 \cdot \vec{r}} e^{-i\omega t}$$

$$\vec{E}_1(\vec{r}, t) = \vec{E}_0 e^{i\vec{k}_1 \cdot \vec{r}} e^{-i\omega t}$$

$$\vec{E}_2(\vec{r}, t) = \vec{E}_0 e^{i\vec{k}_2 \cdot \vec{r}} e^{-i\omega t} \quad \text{导电介质(金属) } \times$$

电磁场边界关系 (绝缘介质下 / 边界无自由电荷、传导电流)

$$\begin{cases} \epsilon_1 E_{1n} = \epsilon_2 E_{2n} & (D_n \text{ 连续}) \\ E_{1t} = E_{2t} \\ \mu_1 H_{1n} = \mu_2 H_{2n} & (B_n \text{ 连续}) \\ H_{1t} = H_{2t} \end{cases}$$

$$\text{入射} \begin{cases} \vec{E}_1 = E_1 e^{i(\vec{k}_1 \cdot \vec{r} - \omega t)} \\ \vec{H}_1 = H_1 e^{i(\vec{k}_1 \cdot \vec{r} - \omega t)} \end{cases}$$

$$\text{反射} \begin{cases} \vec{E}' = E' e^{i(\vec{k}' \cdot \vec{r} - \omega t)} \\ \vec{H}' = H' e^{i(\vec{k}' \cdot \vec{r} - \omega t)} \end{cases}$$

$$\text{折射} \begin{cases} \vec{E}_2 = E_2 e^{i(\vec{k}_2 \cdot \vec{r} - \omega t)} \\ \vec{H}_2 = H_2 e^{i(\vec{k}_2 \cdot \vec{r} - \omega t)} \end{cases}$$

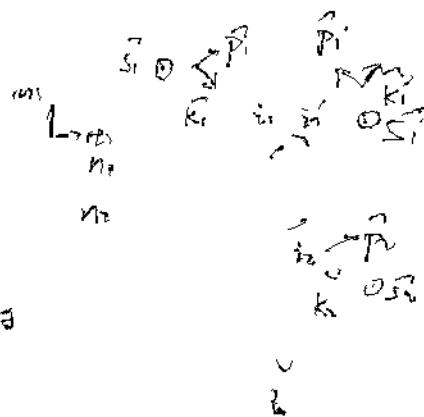
$$\theta, \gamma, \gamma' \text{ 都满足边界关系} \Rightarrow (k_1)_x = (k'_1)_x = (k_2)_x \quad \omega_1 = \omega'_1 = \omega_2 = \omega$$

$$(k_1)_y = (k'_1)_y = (k_2)_y$$

取 xOz 为入射面, 则 $k_{1y} = 0 \Rightarrow k'_{1y} = k_{2y} = 0$ (反、折射光线在平面内)

$$\sin \theta_1 = \frac{k_{1x}}{k_1}, \sin \theta'_1 = \frac{k'_{1x}}{k'_1}, \sin \theta_2 = \frac{k_{2x}}{k_2} \quad \text{且 } k = \frac{\omega}{c}$$

$$\theta_1 = \theta'_1$$



$$n_1 \sin i_1 = n_2 \sin i_2$$

$$\begin{aligned} * k_{12} &= |k_1^2 - k_2^2| = k_2 |1 - \sin^2 i_2| \\ &= k_2 \sqrt{1 - \left(\frac{n_1 \sin i_1}{n_2}\right)^2} = k_1 \sqrt{\left(\frac{n_2}{n_1}\right)^2 - \sin^2 i_1} \\ &= k_1 |\sin^2 i_2 - \sin^2 i_1| \end{aligned}$$

→ $n_1 > n_2$ 时, k_{12} 可能为虚数 ($i_1 > i_c$ 时)

边界关系:

$$\begin{cases} -\epsilon_1 \langle \vec{E}_{1p} \sin i_1 + \vec{E}_{1p}' \sin i_1' \rangle = -\epsilon_2 \langle \vec{E}_{2p} \sin i_2 \rangle \\ \vec{E}_{1p} \cos i_1 - \vec{E}_{1p}' \cos i_1' = \vec{E}_{2p} \cos i_2 \\ -\mu_1 \langle \vec{H}_{1p} \sin i_1 + \vec{H}_{1p}' \sin i_1' \rangle = -\mu_2 \langle \vec{H}_{2p} \sin i_2 \rangle \\ \vec{H}_{1p} \cos i_1 - \vec{H}_{1p}' \cos i_1' = \vec{H}_{2p} \cos i_2 \end{cases}$$

$$\text{解得} \quad \vec{E}_{1p}' = \frac{\epsilon_2 \sin i_2 \cos i_1 - \epsilon_1 \sin i_1 \cos i_2}{\epsilon_1 \sin i_1' \cos i_2 + \epsilon_2 \sin i_2 \cos i_1} \vec{E}_{1p}$$

$$\vec{E}_{2p} = \frac{\epsilon_1 (\sin i_1 \cos i_1' + \sin i_1' \cos i_1)}{\epsilon_1 \sin i_1' \cos i_2 + \epsilon_2 \sin i_2 \cos i_1} \vec{E}_{1p}$$

$$\vec{H}_{1p}' = -\sqrt{\frac{\epsilon_1}{\mu_1}} \vec{E}_{1s}' = \frac{\mu_2 \sin i_2 \cos i_1 - \mu_1 \sin i_1 \cos i_2}{\mu_1 \sin i_1' \cos i_2 + \mu_2 \sin i_2 \cos i_1} \vec{H}_{1p}$$

$$\vec{H}_{2p} = -\sqrt{\frac{\epsilon_2}{\mu_2}} \vec{E}_{2s}' = \frac{\mu_1 (\sin i_1 \cos i_1' + \sin i_1' \cos i_1)}{\mu_1 \sin i_1' \cos i_2 + \mu_2 \sin i_2 \cos i_1} \vec{H}_{1p}$$

↓

菲涅耳公式:

$$\vec{E}_{1p}' = \frac{n_2 \cos i_1 - n_1 \cos i_2}{n_2 \cos i_1 + n_1 \cos i_2} \vec{E}_{1p} \quad \vec{E}_{1p} = \frac{\tan(i_1 - i_2)}{\tan(i_1 + i_2)} \vec{E}_{1p}$$

$$\vec{E}_{2p} = \frac{2n_1 \cos i_1}{n_2 \cos i_1 + n_1 \cos i_2} \vec{E}_{1p}$$

$$\vec{E}_{1s}' = \frac{n_1 \cos i_1 - n_2 \cos i_2}{n_1 \cos i_1 + n_2 \cos i_2} \vec{E}_{1s} \quad \vec{E}_{1s} = \frac{\sin(i_2 - i_1)}{\sin(i_2 + i_1)} \vec{E}_{1s}$$

$$\vec{E}_{2s} = \frac{2n_1 \cos i_1}{n_1 \cos i_1 + n_2 \cos i_2} \vec{E}_{1s} \quad \vec{E}_{1s} = \frac{2 \sin i_2 \cos i_1}{\sin(i_1 + i_2)} \vec{E}_{1s}$$

应用条件:

① 各向同性介质 介电常数 → ϵ

$\vec{E}_1, \vec{E}_2, \vec{E}_1', \vec{E}_2', \vec{H}_1, \vec{H}_2, \vec{H}_1', \vec{H}_2'$

② $r_p = \frac{E_{rp}}{E_{ip}} = \frac{n_2 \cos \theta_i - n_1 \cos \theta_r}{n_2 \cos \theta_i + n_1 \cos \theta_r} = \frac{\tan(\theta_i - \theta_r)}{\tan(\theta_i + \theta_r)}$

③ $r_s = \frac{E_{rs}}{E_{is}} = \frac{n_1 \cos \theta_i - n_2 \cos \theta_r}{n_1 \cos \theta_i + n_2 \cos \theta_r} = \frac{\sin(\theta_r - \theta_i)}{\sin(\theta_r + \theta_i)}$

2.

(1) 反射波和折射波

$$\left\{ \begin{aligned} \tilde{r}_p &= \frac{\tilde{E}_{rp}}{\tilde{E}_{ip}} = \frac{n_2 \cos \theta_i - n_1 \cos \theta_r}{n_2 \cos \theta_i + n_1 \cos \theta_r} = \frac{\tan(\theta_i - \theta_r)}{\tan(\theta_i + \theta_r)} \\ \tilde{r}_s &= \frac{\tilde{E}_{rs}}{\tilde{E}_{is}} = \frac{n_1 \cos \theta_i - n_2 \cos \theta_r}{n_1 \cos \theta_i + n_2 \cos \theta_r} = \frac{\sin(\theta_r - \theta_i)}{\sin(\theta_r + \theta_i)} \end{aligned} \right.$$

$$\left\{ \begin{aligned} \tilde{t}_p &= \frac{\tilde{E}_{tp}}{\tilde{E}_{ip}} = \frac{2n_1 \cos \theta_i}{n_1 \cos \theta_i + n_2 \cos \theta_r} \\ \tilde{t}_s &= \frac{\tilde{E}_{ts}}{\tilde{E}_{is}} = \frac{2n_1 \cos \theta_i}{n_1 \cos \theta_i + n_2 \cos \theta_r} = \frac{2 \sin \theta_r \cos \theta_i}{\sin(\theta_i + \theta_r)} \end{aligned} \right.$$

$$\begin{aligned} \cos \theta_r &= \frac{1}{\sqrt{1 + \tan^2 \theta_r}} \\ &= \frac{1}{\sqrt{1 + \left(\frac{n_1}{n_2}\right)^2}} = \frac{n_2}{\sqrt{n_1^2 + n_2^2}} \end{aligned}$$

$$\cos \theta_i = \sin \theta_r = \frac{n_2}{\sqrt{n_1^2 + n_2^2}}$$

eg. 正入射时 $\theta_i = \theta_r = 0$

$$\tilde{r}_p = \frac{n_2 - n_1}{n_2 + n_1} \quad \tilde{r}_s = \frac{2n_1}{n_1 + n_2}$$

$$\tilde{t}_p = \frac{2n_1}{n_1 + n_2} \quad \tilde{t}_s = \frac{2n_1}{n_1 + n_2}$$

* $\tilde{r}_p < 0$?

振动方向与电场相反

$n_1 < n_2$ 时, $\nearrow \searrow$ 入射, 反射振动方向相反, 差 π
 $\tilde{r}_p > 0$

$n_1 > n_2$ 时, $\nearrow \nearrow$ 无相位差
 $\tilde{r}_p < 0$ ($\tilde{r}_p < 0$)

正入射时, 系统有轴对称性, 但 $\tilde{r}_p \neq \tilde{r}_s$?

$\tilde{t}_p, \tilde{t}_s > 1$?

2) 光强反射 / 透射率

$$R_p = \frac{I_p'}{I_p} = r_p^2 \quad R_s = r_s^2$$

$$T_p = \frac{I_p}{I_p} = \frac{n_2}{n_1} t_p^2 \quad T_s = \frac{I_s}{I_s} = \left[\frac{n_2}{n_1} \right] t_p^2 \quad \rightarrow \quad I \propto n^2 z^2$$

$$R_p + T_p = r_p^2 + t_p^2 = \left(\frac{n_2 \cos i_1 - n_1 \cos i_2}{n_1 \cos i_1 + n_1 \cos i_2} \right)^2 + \left(\frac{2n_1 \cos i_1}{n_2 \cos i_1 + n_1 \cos i_2} \right)^2$$

$$= \frac{(n_2 \cos i_1)^2 + n_1^2 \cos^2 i_2 - 2n_1 n_2 \cos i_1 \cos i_2 + 4n_1^2 \cos^2 i_1}{(n_2 \cos i_1 + n_1 \cos i_2)^2} = 1$$

3) 射流透射率 / 反射率

解法 $W = IS$

$$R_p = \frac{W_p'}{W_p} = R_p \quad R_s = R_s$$

$$T_p = \frac{W_p}{W_p} = \frac{\cos i_2}{\cos i_1} T_p$$

$$T_s = \frac{\cos i_2}{\cos i_1} T_s$$

$$R_p + T_p = \left(\frac{n_2 \cos i_1 - n_1 \cos i_2}{n_2 \cos i_1 + n_1 \cos i_2} \right)^2 + \frac{n_2 \cos i_1}{n_1 \cos i_1} \left(\frac{2n_1 \cos i_1}{n_2 \cos i_1 + n_1 \cos i_2} \right)^2$$

$$= \frac{n_2^2 \cos^2 i_1 + n_1^2 \cos^2 i_2 + 2n_1 n_2 \cos i_1 \cos i_2}{(n_2 \cos i_1 + n_1 \cos i_2)^2} = 1$$

$$\frac{S_1}{S_2} = \frac{\cos i_1}{\cos i_2}$$

$$\frac{i_1}{S_1} = \frac{i_2}{S_2}$$

(4) Brewster 角

$r_p = \frac{\tan(i_1 - i_2)}{\tan(i_1 + i_2)}$ 当 $i_1 + i_2 = \frac{\pi}{2}$ $r_p \rightarrow 0$ 此时入射角为 i_2

$\begin{cases} n_1 \sin i_2 = n_2 \sin i_1 \\ i_2 + i_1 = \frac{\pi}{2} \end{cases}$ 解得 $\tan i_2 = \frac{n_2}{n_1}$

* 此时反射、折射光线成直角

且反射光全为s光

3. Stokes 倒逆关系

叠加后，反射、折射光线偏振

$$\begin{cases} i \rightarrow r \\ i \rightarrow t \end{cases} \Rightarrow \begin{cases} 1 = r^2 + t^2 \\ r t + r' t' = 0 \end{cases}$$

$r^2 + t^2 = 1$

$r t + r' t' = 0$

$r' = -r$

$t t' + r^2 = 1$

验证: Stokes 关系 $\Leftrightarrow$ Fresnel 公式

$\tilde{r}_p = \frac{\tan(i_1 - i_2)}{\tan(i_1 + i_2)} = -\tilde{r}_p' = \frac{\tan(i_1 - i_2)}{\tan(i_1 + i_2)}$

$\tilde{r}_s = \frac{\sin(i_1 - i_2)}{\sin(i_1 + i_2)} = -\tilde{r}_s' = \frac{\sin(i_2 - i_1)}{\sin(i_1 + i_2)}$

$\tilde{r}_p \cdot \tilde{r}_p' + \tilde{r}_p \tilde{r}_p' = \frac{2n_1 \cos i_1}{n_1 \cos i_1 + n_2 \cos i_2} \cdot \frac{2n_1 \cos i_2}{n_2 \cos i_1 + n_1 \cos i_2} + \left(\frac{n_2 \cos i_1 - n_1 \cos i_2}{n_1 \cos i_2 + n_2 \cos i_1} \right)^2$

$= 1$

$\tilde{t}_s \cdot \tilde{t}_s' + \tilde{r}_s \tilde{r}_s' = \frac{2 \sin i_1 \cos i_1}{\sin(i_1 + i_2)} \cdot \frac{2 \sin i_2 \cos i_2}{\sin(i_1 + i_2)} + \left(\frac{\sin(i_2 - i_1)}{\sin(i_1 + i_2)} \right)^2$

$= \frac{4 \sin i_1 \sin i_2 \cos i_1 \cos i_2 + \sin^2 i_1 \cos^2 i_1 + \sin^2 i_2 \cos^2 i_2 - 2 \sin i_1 \sin i_2 \cos i_1 \cos i_2}{\sin^2(i_1 + i_2)}$

$\sin^2(i_1 + i_2)$

$$\sin^2(\theta_1 + \theta_2)$$

例：玻璃窗（透射起偏器）透射光的偏振度

$$\text{偏振度 } P = \frac{I_m - I_m}{I_m + I_m} = \frac{I_p - I_s}{I_p + I_s}$$

为什么 $I_m = I_p$, $I_m = I_s$? $\rightarrow R_s \approx R_p$, $T_s < T_p$

① 空气-玻璃界面.

$$\text{而 } \frac{I_{Ts}}{I_{Tp}} = \frac{t_s^2}{t_p^2} \cdot \frac{I_{is}}{I_{ip}}$$

当入射角为 Brewster 角时自然光入射

$$I_{Ts} = I_{Tp} \quad t_s = 2 \cdot \frac{n_1}{n_2} \cdot \left(\frac{n_1 n_2}{n_1^2 + n_2^2} \right)^2 = \frac{2n_1^2}{n_1^2 + n_2^2} \quad t_p = \frac{n_1}{n_2}$$

$$\text{代入 } \frac{I_{Ts}}{I_{Tp}} = \left(\frac{2n_1 n_2}{n_1^2 + n_2^2} \right)^2 \cdot \frac{I_{is}}{I_{ip}} = \frac{4n_1^2 n_2^2}{(n_1^2 + n_2^2)^2} \cdot \frac{I_{is}}{I_{ip}} = \frac{4n_1^2 n_2^2}{(n_1^2 + n_2^2)^2}$$

$$P_1 = \frac{1 - \frac{I_{Ts}}{I_{Tp}}}{1 + \frac{I_{Ts}}{I_{Tp}}} = \frac{1 - \frac{4n_1^2 n_2^2}{(n_1^2 + n_2^2)^2}}{1 + \frac{4n_1^2 n_2^2}{(n_1^2 + n_2^2)^2}} = \frac{(n_1^2 - n_2^2)^2}{(n_1^2 n_2^2)^2 + 4n_1^2 n_2^2}$$

② 玻璃-空气界面.

$$P_2 = \frac{I_{Tp}' - I_{Ts}'}{I_{Tp}' + I_{Ts}'} = \frac{1 - \frac{I_{Ts}'}{I_{Tp}'}}{1 + \frac{I_{Ts}'}{I_{Tp}'}}$$

$$I = n E^2$$

$$\text{其中 } I_{Tp}' = I_{ip} \quad I_{Ts}' = n_1 \cdot (E_{Ts}')^2 = n_1 (t_s' \cdot E_{is})^2$$

$$= n_1 t_s'^2 E_{is}^2 = \left(\frac{n_1}{n_2} \right)^2 t_s'^2 I_{is}$$

$$= \frac{n_1}{n_2} t_s'^2 \cdot \frac{n_2}{n_1} t_s^2 I_{is} = (t_s' t_s)^2 I_{is}$$

$$t_s = \frac{2n_1^2}{n_1^2 + n_2^2} \quad t_s' = \frac{2n_2^2}{n_1^2 + n_2^2}$$

$$\Rightarrow P_2 = \frac{1 - (t_s' t_s)^2}{1 + (t_s' t_s)^2} = \frac{1 - \left[\frac{4n_1^2 n_2^2}{(n_1^2 + n_2^2)^2} \right]^2}{1 + \left[\frac{4n_1^2 n_2^2}{(n_1^2 + n_2^2)^2} \right]^2}$$

$\Rightarrow$ 经过两块玻璃片后

$$1 - \frac{I_{Ts}'}{I_{Tp}'} \quad 1 - (t_s' t_s)^{2N}$$

$$\begin{matrix} 1 \\ n_1 \\ n_2 \\ n_1 \end{matrix} \quad \begin{matrix} \downarrow \\ \downarrow \\ \downarrow \\ \downarrow \end{matrix} \quad \begin{matrix} 1 \\ \downarrow \\ \downarrow \\ \downarrow \end{matrix}$$

$$P_N = \frac{1 + \frac{2n_1^2}{2n_2^2}}{1 + \frac{2n_1^2}{2n_2^2}} = \frac{1 + \frac{2n_1^2}{2n_2^2}}{1 + \frac{2n_1^2}{2n_2^2}} \quad \text{代入 } n_1=1, n_2=\frac{1}{2}$$

$$P_d = 0.87$$

4. 相位关系和半透损失问题

$$\tilde{r}_p = \frac{n_2 \cos i_1 - n_1 \cos i_2}{n_2 \cos i_1 + n_1 \cos i_2} = r_p e^{i\delta_p} \quad \text{而 } \tilde{t}_i, \tilde{t}_p \text{ 总是正实数}$$

(非全反射)

$$\tilde{r}_s = \frac{n_1 \cos i_1 - n_2 \cos i_2}{n_1 \cos i_1 + n_2 \cos i_2} = r_s e^{i\delta_s}$$

$$\delta_p' = \phi_{p'} - \phi_{ip} \quad \delta_s' = \phi_{s'} - \phi_{is}$$

$$\text{设 } \phi_{ip} = \phi_{is} = 0$$

① 当 $\delta_p' = \delta_s' = 0$ ($\tilde{r}_p, \tilde{r}_s$ 为正实数)

$\Rightarrow$ 反射光振动态与局部坐标架 (p', s') 一致

② 当 $\delta_p' = \delta_s' = \pi$ ($\tilde{r}_p, \tilde{r}_s$ 为负实数)

$\Rightarrow$ 反射光振动态与局部坐标架 (p', s') 相反

③ $\delta_p', \delta_s' \neq 0$

$\Rightarrow$ 反射光振动态介于两者之间

入射为线偏振光, 反射为椭圆偏振光

I. $n_1 < n_2$

$$\begin{cases} \delta_p' = 0 & (i < i_B) & \delta_p' = \pi & (i > i_B) \\ \delta_s' = \pi & (\sin(i_2 - i_1) < 0) \end{cases}$$

II. $n_1 > n_2$

$$\delta_p' = \pi \quad (i < i_B) \quad \delta_p' = 0 \quad (i_B < i < i_c)$$

$$\begin{aligned} n_2 \cos i_1 - n_1 \cos i_2 &= \left(1 - \frac{\sin i_2 \cos i_2}{\sin i_1 \cos i_1}\right) n_2 \cos i_1 \\ &= \frac{\sin(2i_1) - \sin(2i_2)}{\sin(2i_1)} n_2 \cos i_1 \end{aligned}$$

其中 i_c 满足 $n_1 \sin i_c = n_2$ 且 $i_c = \arcsin(\frac{n_2}{n_1}) > \arctan(\frac{n_2}{n_1}) = i_B$

$$i > i_c \text{ 时 } \cos i_2 = \sqrt{1 - (\sin i_2)^2} = \sqrt{1 - \left(\frac{n_1 \sin i_1}{n_2}\right)^2} = i$$

$$\tilde{r}_m = n_2 \cos i_1 \left[1 - \frac{n_1 \sin i_1}{n_2} \right]^{-1} i \quad \text{选取 } e^{-i\omega t} \text{ (相位超前取负, 落后取正)}$$

$$n_2 \cos i_1 \left(+ \right) \frac{n_1}{n_2} \left(\frac{n_1 \sin i_1}{n_2} \right)^2 - 1 \quad ?$$

因此实际相位差是正的

$$\delta_p' = 2 \arctan \left(\frac{n_1}{n_2} \sqrt{\left(\frac{n_1 \sin i_1}{n_2} \right)^2 - 1} \right)$$

$$i_1 \rightarrow \frac{\pi}{2} \quad \delta_p \rightarrow 2 \cdot \arctan(\infty) = \pi$$

$$\delta_s' = 0 \quad (i_1 < i_c)$$

$$\left(n_1 \cos i_1 - n_2 \cos i_2 = \left(1 - \frac{n_2 \cos i_2}{n_1 \cos i_1} \right) n_1 \cos i_1 = \frac{\sin(i_2 - i_1)}{\sin i_1 \cos i_2} \quad n_2 \cos i_2 \right)$$

$$\delta_s' = 2 \arctan \left(\frac{n_2}{n_1} \sqrt{\left(\frac{n_1 \sin i_1}{n_2} \right)^2 - 1} \right) \quad (i_c < i_1 < \frac{\pi}{2})$$

$$i_1 \rightarrow \frac{\pi}{2} \quad \delta_s' \rightarrow 2 \cdot \arctan(\infty) = \pi$$

结论:

1. 正入射

$$\begin{cases} n_1 < n_2 & \text{反射光相位突变 } \pi \\ n_1 > n_2 & \text{无相位突变} \end{cases}$$

2. 掠入射

反射光总有相位突变 π

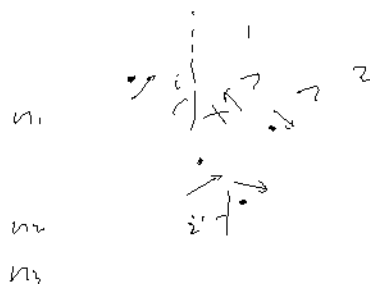
3. 斜入射

在薄膜干涉中:

$i < i_c$ 时

$$\textcircled{1} n_1 < n_2 > n_3 \text{ (或 } n_1 > n_2 < n_3)$$

$$\text{上: } \delta_p' = \pi \quad \delta_s' = 0$$



$$\sin i' = \frac{n_2}{n_1} \sin i < \frac{n_1}{n_2} \sin(i_2) = \frac{n_1}{n_2} \cdot \frac{n_2}{\sqrt{n_1^2 + n_2^2}} = \frac{n_1}{\sqrt{n_1^2 + n_2^2}}$$

$$\tan i' = \frac{n_1}{n_2} < \frac{n_2}{n_3} = \tan(i_3') \Rightarrow i' < i_3'$$

$$\text{下: } \delta_p' = \pi, \delta_s' = 0$$

$$\Rightarrow 1, 2 \text{ 相位差为 } \pi, \Delta L = (\Delta L)_0 \pm \frac{\lambda}{2}$$

$$\textcircled{2} n_1 > n_2 > n_3 \text{ 或 } n_1 < n_2 < n_3$$

$$\Rightarrow 1, 2 \text{ 相位差为 } 0, \Delta L = (\Delta L)_0$$

例: 射电星信号从地球升起

例: 菲涅耳棱镜产生圆偏振光

掠入射 $\Rightarrow$ 总是有半波损

入射光偏振方向与水平面夹角 $\alpha = 45^\circ$, $\varphi_P - \varphi_{rs} = 0$

* 在薄膜干涉中, 上下两束反射光束总有半波损

5. 全反射时的透射场 - 隐失波

$$n_1 < n_2, \quad i_1 > i_c \text{ 即}$$

$$k_{2z} = k_1 \sqrt{\left(\frac{n_2}{n_1}\right)^2 - (\sin i_1)^2} = i k_{2z}'$$

$$\begin{aligned} \vec{E}_2 &= E_2 e^{i(\vec{k} \cdot \vec{r} - \omega t)} = E_2 e^{i(k_{2x} \cdot x + k_{2z} \cdot z - \omega t)} \\ &= E_2 e^{-k_{2z}' z} e^{i(k_{2x} \cdot x - \omega t)} \end{aligned}$$

引入穿透深度 d : $e^{-k_{2z}' d} = \frac{1}{e}$

$$\text{即 } d = \frac{1}{k_{2z}'} = \frac{1}{k_1 \sqrt{(\sin i_1)^2 - \left(\frac{n_2}{n_1}\right)^2}} = \frac{\lambda}{2\pi \sqrt{(\sin i_1)^2 - \left(\frac{n_2}{n_1}\right)^2}} \sim \lambda$$

性质: (1) 等幅面与等相位面正交 (非均匀波)

$$(2) \quad V_x = \frac{\omega}{k_x} < V_2 = \frac{\omega}{k_2} \quad (k_{2x} > k_2)$$

代入 $k_{2x} = k_{1x} = k_1 \sin i_1$

$$\Rightarrow V_x = \frac{V_1}{\sin i_1}, \quad \lambda_x = \frac{2\pi}{k_{2x}} = \frac{\lambda_1}{\sin i_1}$$

3) 射线传播:

$$\vec{S}_e = \vec{E}_x \times \vec{H}_y - \vec{E}_y \times \vec{H}_x$$

$$\langle \vec{S}_e \rangle = 0 \quad (\text{垂直方向无射线流})$$

§2 双折射

1. 各向异性材料

两束线偏振光 $\left\{ \begin{array}{l} o \text{光} \text{ 振动方向 } \perp o \text{光主平面} \\ e \text{光} \end{array} \right. \quad \therefore e \text{光主平面}$

光轴: 不发生双折射方向

主平面: 入射光线 & 光轴 $\rightarrow$ 重合时, o, e 光主平面重合
主平面: 法线 & 光轴 $\rightarrow o, e$ 光振动方向垂直.

2. 单轴 (单轴)

各向异性 — 等相位面为球面 / 非球面 (椭球面) 沿不同方向传播速度不同

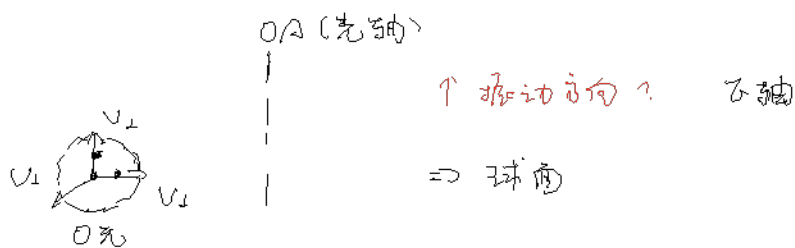
$\downarrow$
 $v \sim$ 振动方向 $\left\{ \begin{array}{l} v_{\perp} = v_e \text{ 振动方向垂直于光轴} \\ v_{\parallel} = v_o \text{ (沿光轴) 平行于} \\ \text{介于 } v_{\parallel}, v_{\perp} \text{ 之间 其他方向} \end{array} \right.$

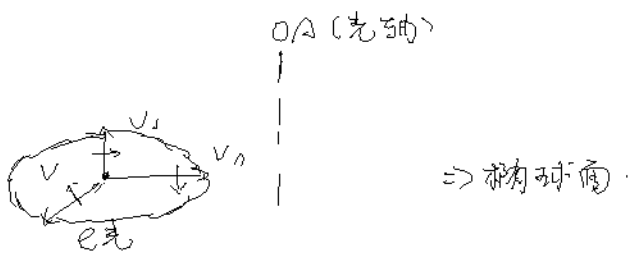
$v_o > v_e$ 正轴晶体 $v_o < v_e$ 负轴晶体

定义 折射率

$$n_o = \frac{c}{v_o} \quad n_e = \frac{c}{v_e}$$

$\Delta n = n_e - n_o$, 反映了双折射的程度 ($\Delta n > 0$ 正, $\Delta n < 0$ 负)





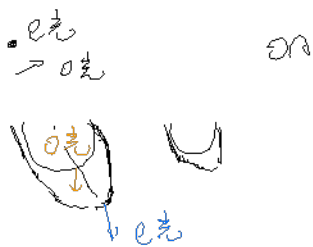
$\Rightarrow$ 椭圆面.

在主截面内斜入射

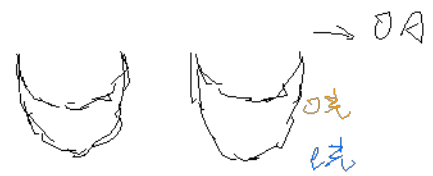
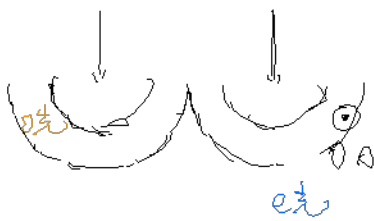
$\rightarrow$ e 光: e 入射时, 折射角不为0, 不遵循折射定律

e 入射,

光轴平行于法线



e 入射: 光轴平行于表面



相对光强

入射面在主截面内时, o, e 光平面重合, o, e 光振动方向垂直

$$\vec{E} = \vec{E}_o + \vec{E}_e = E \cos \theta \hat{j} + E \sin \theta \hat{e}$$

① 自然光入射: $I_o = I_e = \frac{1}{2} I_0$

② 线偏振光入射

取决于 θ

双折射起偏器 — 使 O 光、E 光分开

e.g. Nicol 棱镜 方解石, 只有 e 光出射 (O 光全反射)

Glan-Thompson 棱镜

e 光也透偏析晶体中?

3. 法向 / 射线速度

法向速度: $V_n = \frac{d}{dt}$ 射线速度 $V_r = \frac{c}{n}$

$$V_n = V_r \cos \alpha$$

法向面: $\left(\frac{V_n}{c}\right)^2 = \frac{\cos^2 \theta}{n_o^2} + \frac{\sin^2 \theta}{n_e^2}$ $\leftarrow V_n$ 与光轴夹角

射线面: $\left(\frac{c}{V_r}\right)^2 = n_o^2 \cos^2 \theta + n_e^2 \sin^2 \theta$ $\leftarrow V_r$ 与光轴夹角

$$\begin{aligned} 0: c^2 &= n_o^2 (V_r \cos \theta)^2 + n_e^2 (V_r \sin \theta)^2 \\ &= n_o^2 z^2 + n_e^2 x^2 \end{aligned}$$

$$\frac{dx}{dz} = -\frac{n_o^2}{n_e^2} \frac{z}{x} = \tan(\frac{z}{x} + \theta) = -\cot \theta$$

$$\rightarrow n_o^2 \tan \theta = n_e^2 \tan \theta$$

$$n(\theta) = \frac{c}{V_n} = \frac{n_o n_e}{n_o^2 \sin^2 \theta + n_e^2 \cos^2 \theta}$$

4. 折射率椭球

$$\vec{D} = \epsilon \epsilon_0 \vec{E}$$

各向异性介质中:

$$\vec{D} = \begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \epsilon_0 \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = \epsilon \epsilon_0 \vec{E}$$

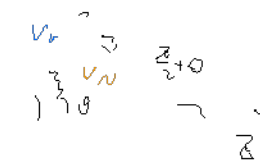
↓ 将 ϵ 对角化

$$D_{x'} = \epsilon_a \epsilon_0 E_{x'}$$

$$D_{y'} = \epsilon_b \epsilon_0 E_{y'}$$

双轴晶体 $\epsilon_a = \epsilon_b + \epsilon_c$

OA



$$D' = \epsilon_c \epsilon_0 E'$$

$$\text{单轴晶体} \quad \epsilon_a = \epsilon_b \neq \epsilon_c$$

5. 双轴晶体

§4 晶体光学器件

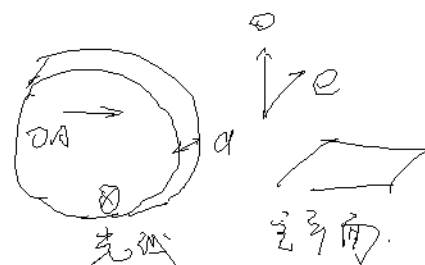
1. 波晶片

$$L_0 = n_o d, \quad L_e = n_e d$$

$$\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta L = \frac{2\pi}{\lambda} (n_o - n_e) d$$

$$\hookrightarrow \text{半波片: } (n_o - n_e) d = \frac{1}{2} \lambda, \quad \Delta = \pi$$

$$\text{四分之一波片: } (n_o - n_e) d = \frac{1}{4} \lambda, \quad \Delta = \frac{\pi}{2}$$



2. 垂直振动的合成

$$\begin{cases} E_x = A_x \cos(\omega t) \\ E_y = A_y \cos(\omega t + \delta) \end{cases}$$

$$\textcircled{1} \delta = 0, \pi$$

$$\delta = 0$$

$$\delta = \pi$$

$$\theta$$

$$\theta_1$$

$$\theta = \arctan \frac{A_y}{A_x}$$

$$\textcircled{2} \delta = \pm \frac{\pi}{2}$$

$$\delta = \frac{\pi}{2}$$

(右旋)

$$\delta = -\frac{\pi}{2}$$

(左旋)

3. (椭圆) 偏振光的获得

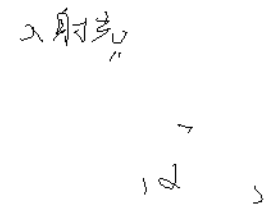
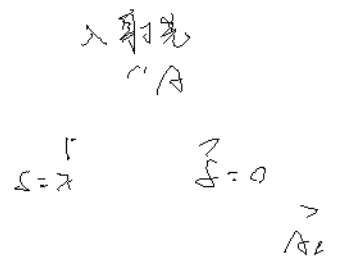
起偏器 + 波晶片

① $\delta_A = \pm \frac{\pi}{2} = \delta_\lambda + A$

↑ 当入射光振动在 -、z $\delta_\lambda = 0$
二、四 $\delta_\lambda = \pi$

$\Rightarrow A = \pm \frac{\pi}{2}$ (1/4 波片)

② $A_0 = A_e$ ($\sin 2 = \cos \alpha \rightarrow \alpha = 45^\circ$)



4. 通过波晶片后偏振状态的变化

① 入射 I

$A_0 = A \sin \alpha, A_e = A \cos \alpha, \delta_\alpha = 0$

② 出射

$\delta_A = \delta_\lambda + A$

实验：圆偏振光 / 自然偏振光，椭圆偏振光 / 部份偏振光



§ 5 偏振光干涉

1. 一般是偏振光干涉

① 入射波晶片 $\delta_{\lambda} = 0/\lambda$

$$A_e = A_1 \cos d \quad A_o = A_1 \sin d \quad \Delta = \frac{2\pi}{\lambda} (n_o - n_e) d$$

② 第 II

$$A_{e2} = A_e \cos \beta$$

$$A_{o2} = A_o \sin \beta \quad \delta_m = 0/\lambda$$

$$A^2 = A_{e2}^2 + A_{o2}^2 + 2 A_{e2} A_{o2} \cos d$$

其中 $\delta = \delta_{\lambda_1} + \Delta + \delta_{\lambda_2}$

例 1 $P_1 \perp P_2$

$$\delta_{\lambda_1} = \pi, \quad \delta_{\lambda_2} = 0$$

$$A_{e2} = A_o = \frac{\sqrt{2}}{2} A_1$$

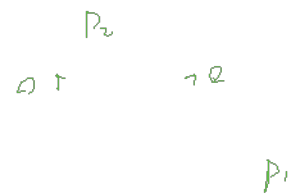
$$A_{e2} = A_{o2} = \frac{1}{2} A_1$$

$$A = \frac{\sqrt{2}}{2} A_1 \sqrt{1 - \cos \Delta}$$

例 2 $P_1 // P_2$

$$\delta_{\lambda_1} = \pi, \quad \delta_{\lambda_2} = \pi$$

$$A = \frac{\sqrt{2}}{2} A_1 \sqrt{1 + \cos \Delta}$$



2. 色偏振

$$\lambda_1 \text{ 满足 } \Delta_1 = \frac{2\pi}{\lambda_1} (n_o - n_e) d = 2k\pi$$

$$\text{则 } P_1 \perp P_2 \text{ 时 } \Delta = 0$$

$$\lambda_2 \text{ 满足 } \Delta_2 = \frac{2\pi}{\lambda_2} (n_o - n_e) d = (2k+1)\pi$$

$$\text{则 } P_1 \parallel P_2 \text{ 时 } \Delta = 0$$

有光入射:

$\Delta = 0 \rightarrow$ 显示互补色

5. 电光效应

1) Kerr 效应

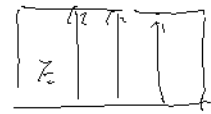
$$\Delta n = n_o - n_e \propto E^2$$

$$\Delta = \frac{2\pi}{\lambda} \cdot \Delta n \cdot d \propto \frac{E^2 d}{\lambda}$$

$$\frac{\Delta}{2\pi} = B \cdot \frac{E^2 d}{\lambda} \quad B \text{ 为物质的 Kerr 常量}$$

2) Pockels 效应

P_1



P_2

偏振片 $P_1 \perp P_2$

§ 6 旋光

1. 旋光现象

$$\psi = \alpha d$$

α 为材料的旋光率

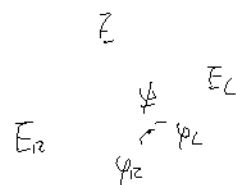
2. 解释.

线偏振光 分解 $\rightarrow$ 左旋圆偏振光 (L光) + 右旋圆偏振光 (R光)

在旋光物质中:

$$v_L \neq v_R \quad \varphi_L = \frac{2\pi}{\lambda} n_L d \quad \varphi_R = \frac{2\pi}{\lambda} n_R d$$

$$\psi = \frac{1}{2} (\varphi_R - \varphi_L) = \frac{\pi}{\lambda} (n_R - n_L) d$$



3. 费里振动

4. 量糖术

$$\psi = [\alpha] d \quad [\alpha] \text{ 为比旋光率.}$$

5. 法拉第效应

振动面旋转 ψ

$$\psi = V B L \quad V \text{ 为 Verdet 常数}$$

($V > 0$ 右旋, $V < 0$ 左旋)



和自然旋光物质区别:

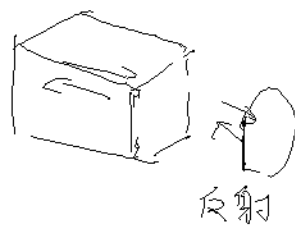
旋光方向与光传播方向无关 (仅与磁场方向有关)

→ Faraday 效应的那更易旋光特性

作用: 光隔离器 (单向导光)

光调制器

0/24



* 磁光 Kerr 效应

线偏振光在处于磁场中的磁性介质表面反射后,

反射光成为椭圆偏振光